

Mind Space — Backend Schema

Document 05 of 06 | Version 1.0 | PFE Project

Prerequisites

- Enable pgvector extension: Dashboard → Database → Extensions → pgvector
- Enable RLS on every table immediately after creation
- Run all migrations in sequence — each depends on the previous

Table: profiles

Auto-created by Postgres trigger on auth.users insert

Column	Type	Constraints	Notes
id	uuid	PK, references auth.users(id) ON DELETE CASCADE	Same UUID as auth user
display_name	text	nullable	User-set display name, defaults to null
created_at	timestamptz	NOT NULL, DEFAULT now()	
updated_at	timestamptz	NOT NULL, DEFAULT now()	Updated by trigger on row change
onboarding_completed	boolean	NOT NULL, DEFAULT false	Set true after onboarding last slide

RLS policies — profiles

- SELECT: auth.uid() = id (users read own profile only)
- UPDATE: auth.uid() = id (users update own profile only)
- INSERT: blocked — handled by trigger only

Trigger — auto-create profile

```
CREATE OR REPLACE FUNCTION handle_new_user()  
RETURNS trigger AS $$  
BEGIN  
    INSERT INTO public.profiles (id)  
    VALUES (NEW.id);  
    RETURN NEW;  
END;  
$$ LANGUAGE plpgsql SECURITY DEFINER;
```

```
CREATE TRIGGER on_auth_user_created
AFTER INSERT ON auth.users
FOR EACH ROW EXECUTE FUNCTION handle_new_user();
```

Table: emotion_spirits

Seeded once. Read-only. No RLS needed (public reference data).

Column	Type	Constraints	Notes
id	integer	PK	1–6 matching app constants
name	text	NOT NULL, UNIQUE	anxious, calm, frustrated, sad, hopeful, overwhelmed
color_hex	text	NOT NULL	Hex code matching design tokens
animation_type	text	NOT NULL	fast_pulse, slow_breath, medium_pulse, etc.

Table: chats

Column	Type	Constraints	Notes
id	uuid	PK, DEFAULT gen_random_uuid()	
user_id	uuid	NOT NULL, FK → profiles(id) ON DELETE CASCADE	
arc_id	uuid	nullable, FK → arcs(id) ON DELETE SET NULL	Populated after end-chat runs assign-arc
title	text	nullable	Derived from first message, not shown in v1
status	text	NOT NULL, DEFAULT active	active completed abandoned
started_at	timestamptz	NOT NULL, DEFAULT now()	
ended_at	timestamptz	nullable	Set when Wrap Up is tapped
message_count	integer	NOT NULL, DEFAULT 0	Incremented by trigger on messages insert

RLS policies — chats

- ALL operations: auth.uid() = user_id

Table: messages

Column	Type	Constraints	Notes
id	uuid	PK, DEFAULT gen_random_uuid()	
chat_id	uuid	NOT NULL, FK → chats(id) ON DELETE CASCADE	
user_id	uuid	NOT NULL, FK → profiles(id) ON DELETE CASCADE	Denormalized for RLS performance
role	text	NOT NULL	user assistant
content	text	NOT NULL	Full message text
created_at	timestamptz	NOT NULL, DEFAULT now()	
flagged_for_safety	boolean	NOT NULL, DEFAULT false	Set true if safety classifier returns CRISIS

RLS policies — messages

- ALL operations: auth.uid() = user_id

Table: reflections

Column	Type	Constraints	Notes
id	uuid	PK, DEFAULT gen_random_uuid()	
user_id	uuid	NOT NULL, FK → profiles(id) ON DELETE CASCADE	
chat_id	uuid	NOT NULL, FK → chats(id) ON DELETE CASCADE	One reflection per chat
arc_id	uuid	nullable, FK → arcs(id) ON DELETE SET NULL	Populated by assign-arc Edge Function
spirit_id	integer	NOT NULL, FK → emotion_spirits(id)	EFT-informed: represents the LAYER beneath surface
what_sage_heard	text	NOT NULL	One sentence: the deeper emotional truth, not the surface label
question_to_sit_with	text	nullable	Optional open question, no answer expected

Column	Type	Constraints	Notes
shared_perspective	text	nullable	"Many people find..." — normalizing community framing
moment_of_clarity	text	nullable	The insight moment if one occurred this session
embedding	vector(1536)	nullable	text-embedding-3-small vector of what_sage_heard
needs_arc_assignment	boolean	NOT NULL, DEFAULT false	True if embedding API failed; for background retry
processing_stage_at_creation	text	nullable	Arc stage at time of this reflection (forming/venting/etc.)
created_at	timestampz	NOT NULL, DEFAULT now()	

RLS policies — reflections

- ALL operations: auth.uid() = user_id

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```
CREATE INDEX reflections_user_embedding_idx
ON reflections USING ivfflat (embedding vector_cosine_ops)
WITH (lists = 100)
WHERE embedding IS NOT NULL;
```

Table: arcs

Column	Type	Constraints	Notes
id	uuid	PK, DEFAULT gen_random_uuid()	
user_id	uuid	NOT NULL, FK → profiles(id) ON DELETE CASCADE	
name	text	NOT NULL	LLM-generated 2–4 word name. User can override.
user_renamed	boolean	NOT NULL, DEFAULT false	If true: LLM never renames this Arc again
dominant_spirit_id	integer	FK → emotion_spirits(id)	Most frequent spirit across sessions

Column	Type	Constraints	Notes
color_hex	text	NOT NULL	Assigned from emotion palette on creation
centroid_embedding	vector(1536)	nullable	Running average of all member reflection embeddings
session_count	integer	NOT NULL, DEFAULT 1	Incremented on each new assignment
processing_stage	text	NOT NULL, DEFAULT forming	forming venting stabilizing processing shifting integrating
status	text	NOT NULL, DEFAULT active	active archived
archived_at	timestampz	nullable	
last_session_at	timestampz	NOT NULL, DEFAULT now()	Updated on each new assignment
created_at	timestampz	NOT NULL, DEFAULT now()	

RLS policies — arcs

- ALL operations: auth.uid() = user_id

Table: arc_insights

Column	Type	Constraints	Notes
id	uuid	PK, DEFAULT gen_random_uuid()	
arc_id	uuid	NOT NULL, FK → arcs(id) ON DELETE CASCADE	
user_id	uuid	NOT NULL, FK → profiles(id) ON DELETE CASCADE	
how_it_evolved	text	NOT NULL	One paragraph on emotional evolution. Never clinical.
pattern_noticed	text	NOT NULL	One observation about a recurring element.
session_count_at_generation	integer	NOT NULL	Snapshot of session_count when generated
generated_at	timestampz	NOT NULL, DEFAULT now()	

Caching logic (in Edge Function, not DB)

- If an insight exists for this arc AND generated_at > (now - 7 days) AND arc.session_count = insight.session_count_at_generation → return cached insight
- Otherwise generate fresh and insert new row (keep history)

RLS policies — arc_insights

- ALL operations: auth.uid() = user_id

Relationships summary

From	To	Type	Via
profiles	chats	one-to-many	chats.user_id
chats	messages	one-to-many	messages.chat_id
chats	reflections	one-to-one	reflections.chat_id
reflections	arcs	many-to-one	reflections.arc_id
arcs	arc_insights	one-to-many	arc_insights.arc_id
reflections	emotion_spirits	many-to-one	reflections.spirit_id
arcs	emotion_spirits	many-to-one	arcs.dominant_spirit_id